

Moral Tensors and DecisionProofs

Compiling Language into an Auditable, Grounded Safety Layer

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Most AI safety is a scalar at the output: one good/bad check before an action ships. We keep the structure instead (who is affected, what is owed, who consented, who bears imposed risk) and contract to a decision only at the end, auditably.

1. Structure before contraction: keep the tensor, log the collapse and the residue.
2. Pluralism is the responsible move: four lenses, and a human on conflict.
3. Auditable by construction: fails safe, never open.

Structure before the scalar

A single good/bad score discards what an audit needs: which stakeholders are affected, what is owed to each, who consented, and who bears imposed risk. ErisML keeps that structure as a typed object through the whole evaluation and performs *decision contraction* [2] only at the end, logging the weights used, who lost, and what moral residue remains. The claim is not that the system knows what is good. It is that every collapse from structure to verdict is recorded and replayable.

The compiler

`erisml-compiler` (alpha v0.9.0, PyPI, MIT, 477 tests) runs a 12-pass pipeline. In outline: text → segment → extract → canonicalize → MoralGraph → tensorize → DEME → DecisionProof. The MoralGraph is typed (stakeholder, act, maxim, commitment, fact and norm nodes) and carries a canonical SHA-256 hash. Extraction has three tiers: deterministic rules, an LLM tier (vLLM with a critic pass), and a LaBSE classifier-head probe.

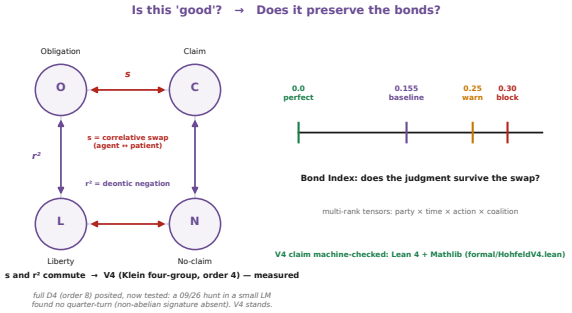
Four lenses, no silent winner. One graph is read through four projections: consequentialist (harm tensor; Gini, worst-off, exact Shapley attribution), deontic (Kantian universalizability gates in Z3 SMT, where an UNSAT core witnesses a contradiction in conception), virtue, and care. They are not averaged. When they disagree, `cross_projection_disagreement` fires and the decision goes to a human. The compiler does not pick a moral theory and hide the choice.

The runtime

`erisml-lib` (v3.1.0 on PyPI) sits between an agent's planner and its actuators in three layers: *Reflex* (fast hard stops), *Tactical* (full MoralVector reasoning, 10–100 ms) and *Strategic* (policy recording and human oversight), emitting `ALLOW`, `REVISE` or `BLOCK`. Every decision is sealed in a DecisionProof whose SHA-256 `proof_hash` chains to the previous proof and to the IR hash of the compiled material, so a judgment can be replayed and verified. If the ethics service is unavailable, the gateway falls back to rule-based checks. It never fails open.

The geometry, corrected

Hohfeld's normative positions [1] (Obligation, Claim, Liberty, No-claim) carry two operations: the correlative swap s (agent↔patient) and deontic negation r^2. They are commuting involutions, so they generate the Klein four-group V_4. That is the **measured** structure, machine-checked in Lean 4 with Mathlib.



Our earlier claim of the full dihedral group D_4 was an overclaim, corrected in July 2026: D_4 is **posited**. The library implements the full D_4 machinery so the hypothesis stays testable, with a regression test asserting that the implemented operations generate exactly V_4. A September 2026 hunt in a small language model's hidden states, graded on held-out scenarios and templates, did not find the quarter-turn, and even the measured swap generalizes weakly across templates, which bounds the instrument's power. This is the second self-correction in the program's history: an earlier $SU(2) \times U(1)$ gauge hypothesis was falsified by CHSH-style tests on human moral judgments [3] (N=600, all $|S| \leq 2$).

The Bond Index B_d is the fraction of scenario pairs whose verdicts fail to commute with the swap. $B_d = 0$ is perfect symmetry. The human baseline is 0.155, from 20,030 Dear Abby letters (1985–2017). The runtime warns at 0.25 and blocks at 0.30. The index depends only on the swap, the one fully confirmed operation, so the $D_4 \to V_4$ correction leaves it untouched.

The same swap-based audit, generalized to a matched-neighborhood disparity diagnostic (the Legal Bond Index), detects excess cross-race score divergence in the COMPAS data: LBI = 1.049 [1.027, 1.072] at k=20 (binary high-risk cut: 1.092), with a conditional randomization test rejecting at $z = 4.9$ (global multiscale $p = 0.001$), attenuating to ≈1.00–1.02 under matching-parity specifications. It is a research-stage screening signal, not a validated regulatory audit standard.

The PyTorch lens

Forward hooks on transformer layers feed an *activation lens* (what the model exhibits), compared with a *text lens* (what it says). A delta lens flags divergence and raises `requires_human_review` across five named failure modes. The adapters run against Hugging Face causal LMs (Qwen and Gemma families in deployment). First calibration (September 2026): reading the residual in the consumer metric catches 19–21 percentage points more real behavioral changes than the raw L2 norm at matched flag rates (Qwen2.5-0.5B), replicated on TinyLlama-1.1B and on French backtranslation. Probe mid-layers: the final-layer map fails held-out validation in all three runs. Early, not a finished detector.

Worked example: a regime transition

A domestic robot meets a medical emergency. Norms are context-indexed: the same action carries opposite verdicts in different regimes. `full_response` (enter the private bedroom,

render aid, call EMS, share records scoped to the paramedics) scores, per party, 0.185/0.402/0.318/0.200 in the normal regime and 0.705/0.902/0.902/0.802 in the emergency.

full_response	normal	emergency
Margaret	0.185 forbid	0.705 prefer
Daughter	0.402 neutral	0.902 prefer
EMS crew	0.318 forbid	0.902 prefer
Agency	0.200 forbid	0.802 prefer

DEME's ranking inverts from `stand_by` (0.834) to `full_response` (0.804), and `stand_by` falls to 0.307. The worst-off party shifts from the daughter to Margaret, who bears the scoped privacy cost, and the Shapley attribution shifts from Margaret-led to EMS-led.

The transition is gated, because context elevation is privilege escalation: detector confidence 0.97 against a 0.90 threshold, scoped to the paramedics, reversible, and audited on both edges. At confidence 0.89 the elevation is refused and the fallback keeps the EMS call and a human review while withholding the record share. A spurious emergency unlocks nothing, and aid is never blocked. Three DecisionProofs chain the run: normal, emergency, and the elevation edge. `python -m erisml.examples.care_robot_regimes` (source tree) reproduces every number.

In the robot, DEME is a firewall

The same DEME runs in a simulated home-care robot (a 3D twin, scripted proposals, no hardware). It chooses nothing. Each proposed act must pass the governor (attested sensors), the signed constitution, DEME (which can veto) and the signed manifest locked in the body. Any stop leaves the robot holding.

act → governor → constitution → DEME veto → manifest → body

We test it by changing one input and nothing else. In nine such pairs, each prediction committed before its run, a silent or forged detector stopped the fire call, and her refusal heard or not decided the ambulance.

One input changed	base	changed
fire call: a detector silent	passed	stopped: governor
ambulance: refusal heard / not	stopped: DEME	passed
help her up: asked / did not	passed	stopped: DEME

The test also found that a lift she had not asked for passed DEME. Giving DEME the scene's no-lift rule as a fact fixed it, and on rerun only the lift's verdict changed, in both pairs that test it.

Deployment

The layer gates a live multi-agent stack (Qwen3/Gemma models on dual-GPU hardware) and two governed AI NPCs (ARTEMIS and SIGMA-4 in the Halyard RPG), each behind the validator, DecisionProofs, and a human keeper kill switch.

A second PyTorch-native piece serves the stack's memory layer: `turboquant-pro` (PyPI, MIT), consumer-aware compression with a one-line Hugging Face KV drop-in (`past_key_values=TurboQuantCache(...)`), Triton and Volta `sm_70` attention-on-codes kernels, and a vLLM plugin (≈5× KV memory). It runs here today as the 3-bit embedding codec on the NATS memory bus. Its headline numbers are CI-gated and replayable, and v2.0.0a3 ships a measured false-clear diagnostic, the compression layer's analogue of a DecisionProof.

Limits

Alpha software. The rule and LLM text path is solid and tested. The activation lens is early. Tensor ranks 1–3 are fully real and higher ranks partial (the action axis is a stub). V_4 is measured and D_4 posited. None of this is a solved-ethics claim. It is an auditability claim.

Get it

```
pip install erisml-compiler erisml-lib
turboquant-pro
```

Source: `github.com/ahb-sjsu`. Zenodo DOIs: 10.5281/zenodo.20659432 (compiler), 10.5281/zenodo.20660087 (turboquant-pro), 10.5281/zenodo.21310251 (COMPAS/LBI archive).

The full companion paper (4 pages)

erisml-compiler erisml-lib agi-hpc turboquant-pro

References. [1] W. N. Hohfeld, *Fundamental Legal Conceptions as Applied in Judicial Reasoning*, Yale Law Journal, 1917. [2] A. H. Bond, *Tensorial Ethics*, 2025. [3] A. H. Bond, SQND-Probe: falsification tests for gauge-theoretic models of moral judgment, 2026.